

Unit-I

Syllabus : Diversity of Organism and Cell Biology: Introduction: Living systems, Bio-mimicry, Metabolism, Taxonomy, Concept of species, Structural organization of life, Concepts of modern cell, history of cell,

Cell theory, cell:- Cell shape, size and cell number, Types of cells:- Prokaryotic cells and Eukaryotic cells, Chemistry of cells. Cell Division: Cell cycle, mitosis, meiosis, mitotic cell division, cell cycle check points, meiotic cell division, embryonic cell division, cell death.

Q. What is Diversity in Living Organisms?

Biodiversity is used to define the diversity of life forms worldwide. It is a word that is used more often to refer to the classification of living species found in a particular geographic region. The Diversity of living species of a geographic region in an area provides stability in the respective region.

There are numerous living organisms on earth with different sizes, shapes, habitats, nutrition, reproduction, and more. That depends on their physical features and their habitat. Animals of any kingdom are classified into different orders and classes.

Animals live in different climates like water, land, grasslands, deserts, forests, ice, water, and ice to forests, deserts, and grasslands. All these organisms consist of cells.

Cells are one of the essential characteristics of living organisms. Cells are structural units of life. It carries out specifically assigned functions in living species. In this way, a group of cells from tissue in living species.

Diversity in living organisms can be seen everywhere on earth. The region of the earth is highly diverse and is called the region of mega biodiversity. Twelve countries in the world have more than half of the biodiversity in the world. India is also one of them.

Over millions of years, diversity has been going on in living beings. Species have evolved from ape-like beings to homo sapiens. People look for similarities between organisms to classify them, and hence they study them as a whole. Regarding this, fundamental characteristics need to be decided, which would form the foundation for classifying.

Introduction to Diversity in Living Organisms

Life exists in different forms on Earth. When it comes to the question of the number of living organisms found on the earth, the answer is unimaginable. This is so because of the large diversity of organisms continuously evolving into a different variety ever since the origin of life had taken place. Diversity is present at different levels like genetic diversity, species diversity, and ecological diversity. Mango alone has around 10,000 varieties in India. This alone example indicates how large and diverse are the living organisms. Gaining knowledge about this large diversity is impossible without classifying them. Thus classification becomes an important step towards the study of different organisms found on the earth.

Metabolism: "Metabolism refers to a series of chemical reactions that occur in a living organism to sustain life." Metabolism is total sum of all biological reactions occurring in any living cell, which are absolutely controlled by enzymes.

There are two categories of metabolism: **catabolism and anabolism.**

Catabolism is the breakdown of organic matter (breaking of molecules (catabolism))

Anabolism is making of new molecules (anabolism) uses energy to construct components of cells, such as proteins and nucleic acids.

Metabolism is the chemical reactions in the body's cells that change food into energy. Our bodies need this energy to do everything from moving to thinking to growing. Specific proteins in the body control the chemical reactions of metabolism.

The three main functions of metabolism are:

The conversion of the energy in food to energy available to run cellular processes

The conversion of food to building blocks for proteins, lipids, nucleic acids, and some carbohydrates The elimination of metabolic wastes.

Difference Between Catabolism and Anabolism	
Catabolism	Anabolism
Catabolism breaks down big complex molecules into smaller, easier to absorb molecules.	Anabolism builds molecules required for the body's functionality.
The process of catabolism releases energy.	Anabolic processes require energy.
Hormones involved in the processes are adrenaline, cytokine, glucagon, and cortisol.	Hormones involved in the process are estrogen, testosterone, growth hormones and insulin.
Examples of catabolic processes are proteins becoming amino acids, glycogen breaking down into glucose and triglycerides breaking up into fatty acids.	Examples include the formation of polypeptides from amino acids, glucose forming glycogen and fatty acids forming triglycerides.
In catabolism, potential energy is changed into kinetic energy.	In anabolism, kinetic energy is converted into potential energy.
It is required to perform different activities in living entities.	It is required for maintenance, growth, and storage.
catabolism, e.g., cell respiration	synthesising reactions (anabolism, e.g., photosynthesis)

Taxonomy: Taxonomy means classification following certain rules or principles. Word Taxonomy comes from two Greek words, taxis – meaning arrangement and nomos meaning law or rule.

Taxonomy is a science that deals with naming, describing and classification of all living organisms including plants.

While categorising the organism, the relationship between each organism will make them get added in a particular group or category. Hence the organisms with similar characteristics will get into the same group., which means they closely resemble each other.

Each category is called a taxon and these taxa are arranged in different levels.

The framework by which taxonomic groups or taxons are arranged in definite order, from lower to higher categories or in progressively complex categories is called taxonomic hierarchy. It is also called the Linnaean hierarchy because it was first proposed by **Linnaeus**.

Taxonomic hierarchies are based on several things such as the domain, kingdom, phylum, class, order, family, genus, and species. Thus, a certain organism is classified on all of these different levels.

For example, human beings can be classified on these different levels, and then, they are allotted a certain taxonomical rank. The scientific classification of human beings is as follows.

- Domain: Eukaryota
- Kingdom: Animalia
- Phylum: Chordata
- Class: Mammalia
- Order: Primates
- Family: Hominidae
- Genus: Homo
- Species: sapiens

Domain

It is the highest rank of organisms and also the most general. It is a much newer rank and it was not developed by the original inventor of the system, that is, Carolus Linnaeus, and it was used for the first time in 1990. The original system was developed in 1735. The three domains of life are bacteria, Eukaryota, and archaea.

Kingdom

It is the rank that is the highest after Domain. Before Domain was invented, Kingdom was the highest rank in the taxonomical hierarchy. There are 7 different kingdoms, which are Animalia, Plantae, Fungi, Protozoa, Chromista, Bacteria, and Archaea.

Phylum

It is the next rank after kingdom and there are 35 phyla in the Animalia class. It is more specific than Kingdom but it is also less specific than class.

Class

It is more specific than phylum but less specific than order. There are 108 different Classes in the Kingdom Animalia.

Order

It is more specific than Class but less specific than Family. Many of Carolus Linnaeus's Orders are used even today and there are 19-26 orders of Mammalia.

Family

It is even more specific and it is less specific than Genus. There are around 12 total families in the Order Carnivora.

Genus

It is even more specific than Family and it is the first part of an organism's scientific name using the binomial nomenclature, with the second part being the species name.

Species

It is the last rank and is the most specific. According to studies, there are around 8.7 million different species of living organisms on Earth. A lot of living organisms are yet to be discovered and categorized.

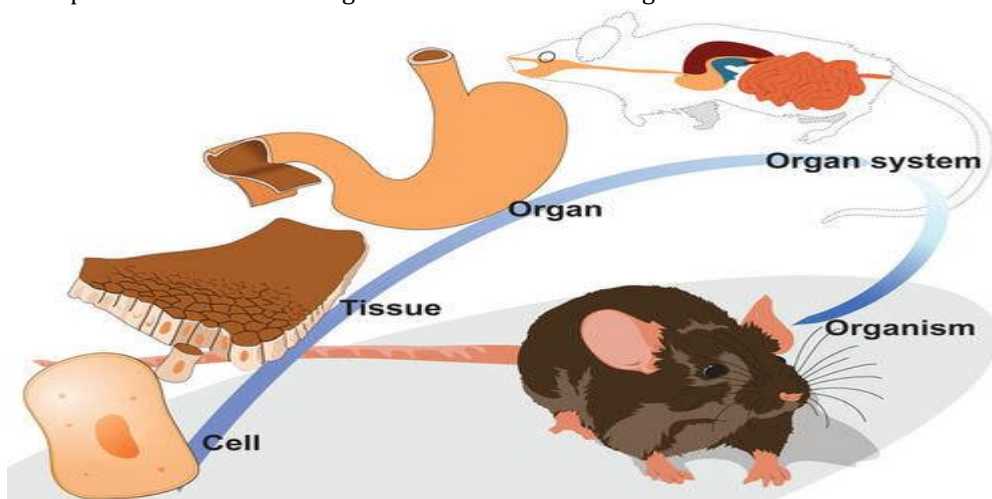
Benefits of taxonomy

There are many benefits for taxonomic studies. Some of them are as follows:

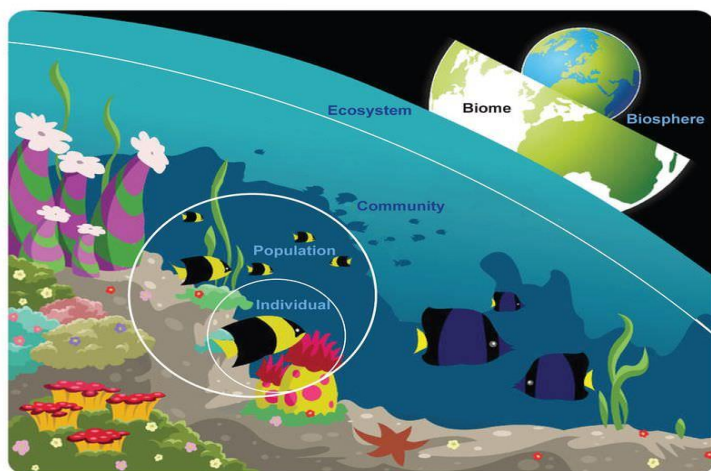
- Categorisation of organisms for ease of study.
- Compares similarities and differences between organisms.
- Shows evolutionary relationships between organisms.
- The same name is used for an organism worldwide.
- Gives us a knowledge of bioresources and their diversity.

Q. Explain Structural Organization of Life (Living Things):

- We know it all starts with the cell. And for some species it ends with the cell. But for others, the cells come together to form tissues, tissues form organs, organs form organ systems, and organ systems combine to form an organism.
- Levels of Organization
- The living world can be organized into different levels. For example, many individual organisms can be organized into the following levels:
- Cell: Basic unit of structure and function of all living things.
- Tissue: Group of cells of the same kind.
- Organ: Structure composed of one or more types of tissues. The tissues of an organ work together to perform a specific function. Human organs include the brain, stomach, kidney, and liver. Plant organs include roots, stems, and leaves.
- Organ system: Group of organs that work together to perform a certain function. Examples of organ systems in a human include the skeletal, nervous, and reproductive systems.
- Organism: Individual living thing that may be made up of one or more organ systems.
- Examples of these levels of organization are shown in Figure below.



- An individual mouse is made up of several organ systems. The system shown here is the digestive system, which breaks down food into a form that cells can use. One of the organs of the digestive system is the stomach. The stomach, in turn, consists of different types of tissues. Each type of tissue is made up of cells of the same type.
- There are also levels of organization above the individual organism. These levels are illustrated in Figure below.
- Organisms of the same species that live in the same area make up a population. For example, all of the goldfish living in the same area make up a goldfish population.
- All of the populations that live in the same area make up a community. The community that includes the goldfish population also includes the populations of other fish, coral, and other organisms.
- An ecosystem consists of all the living things (biotic factors) in a given area, together with the nonliving environment (abiotic factors). The nonliving environment includes water, sunlight, soil, and other physical factors.
- A group of similar ecosystems with the same general type of physical environment is called a biome.
- The biosphere is the part of Earth where all life exists, including all the land, water, and air where living things can be found. The biosphere consists of many different biomes.



- This picture shows the levels of organization in nature, from the individual organism to the biosphere.

Q. Explain in detail Concepts of Species :

The following points highlight the four important groups of species concept. The groups are: 1. Typological Species Concept 2. Nominalistic Species Concept 3. Evolutionary Species Concept 4. Biological Species Concept.

Group # 1. Typological Species Concept:

This concept says that the observed diversity of the universe reflects the existence of a limited number of underlying “universals” or types. Individuals are considered to be merely expressions of the same type. When two individuals or groups of individuals have sufficiently different characters then only they should be considered as different species.

Variation results in the revealing of characters which are already present but not expressed in each species. Thus, variation according to the typological species concept is considered to be an irrelevant phenomenon.

This concept dates back 2,300 years to the philosophies of Plato and Aristotle, and was the species concept of Linnaeus and his followers. Species was supposed to be a fixed or static unit which did not change and, as such, existed permanently and forever.

Since this philosophical tradition is sometimes referred to as essentialism, the typological definition is also sometimes called the essentialist species concept. The species can be recognised by their essential natures or essential characters which are expressed in accordance with their morphology.

It is, therefore, also called the morphological species concept. Taxonomists almost unanimously accepted the essentialist species concept up to the early post-Linnaean period.

It thus includes four postulates:

- (i) Species are similar individuals sharing the same essence.
- (ii) Each species is separated from all others by a sharp discontinuity.
- (iii) Each species is completely constant through time.
- (iv) Strict limits are present to the possible variation within any one species.

Two practical reasons exist for the present universal rejection of this concept:

1. Individuals are frequently found in nature that are clearly conspecific with other individuals in spite of striking differences in structure owing to sexual dimorphism, age differences, polymorphism and other terms of individual variation. These were often described originally as different species.

For example:

- (i) The male and female of river duck, the mallard, were originally placed under separate species. The males were described as *Anas boschas* and the females as *Anas platyrhynchos*.
- (ii) In many groups of birds (humming bird, pea-hen etc.) females differ more from the males of their own species than from the females of other related species.
- (iii) Larval stages of several chordates and invertebrates are markedly different from their parents or adult stage and, thus, were very often considered as separate species.
- (iv) In case of the deep sea fishes the males are dwarf and attached to the body of the females. So they were considered as separate species.

Thus, in the above cases, they should be deprived of their separate species status, regardless of their degree of morphological differences, as soon as they are found to be members of the same breeding population. Different phases that belong to the same population cannot be considered as different species.

2. Sibling species differ hardly at all morphologically, yet are good biological species. Degree of difference is not the decisive criterion in the ranking of taxa as species. The typological species concept is still defended by a few writers.

In situations where there is a lack of biological information, a taxonomist may be forced to recognise a species provisionally on the basis of morphological evidences, but such species are subject to later reconsideration.

Group # 2. Nominalistic Species Concept:

Nominalistic species concept is that of Occam and his followers, who believed that nature produces individuals only. Species are man's own creation and have no actual existence in nature. They are mental concepts and nothing more. Species have been invented so that we may refer to great numbers of individuals collectively. This concept was popular in France during the 18th century.

The drawbacks of this concept are:

- (i) No naturalist whether a primitive native or a trained population geneticist – can agree that species are man-made, when it is an established fact that they are the products of evolution.
- (ii) Nominalists misinterpreted the relation between similarity and relationship. The members of any species are not grouped together as they are similar (as claimed by these workers), rather they are similar to each other because of common heritage. It is just like when two brothers are identical twins not due to their similarity but rather due to both being derived from a single zygote.

Group # 3. Evolutionary Species Concept:

The shortcomings of Biological species concept (in uniparental organisms where interbreeding fails), had led Meglitsch (1954), Simpson (1961), Grant (1971) and other authors – particularly paleontologists – in formulating the evolutionary species concept.

Simpson (1961) defined it as “an evolutionary species is a lineage (an ancestral- descendant sequence of populations) evolving separately from others and with its own unitary evolutionary role and tendencies”. Willey (1981), on the other hand, believed that each species is an internally similar part of a phylogenetic tree.

The drawbacks of this concept are:

(i) This definition is of a phyletic lineage and not of a species. It side-stepped the crucial role of why phyletic lines do not interbreed with each other.

(ii) This concept ignores the core of the species problem as to the causation and maintenance of discontinuities between contemporary species.

(iii) This concept has failed to solve the problem of how to deal with the relationship of descendant populations in a single lineage.

Group # 4. Biological Species Concept:

The biological species concept is also known as the Newer Species Concept, because it was accepted in the latter half of the nineteenth century after Darwin’s “Origin of Species” was published (in 1859) and also due to the fact that organic evolution was established.

It was after 1750 that an entirely new species concept began to emerge. But it was in 1905 that K. Jordan first clearly formulated the concept in all of its consequences. This concept combined the thoughts of the typological and nominalistic concepts by stating that the species have independent reality and are typified by the statistics of populations of individuals.

It, however, differs from both by stressing the populational aspect and genetic cohesion of the species and also by pointing out that species receives its reality from the historically evolved, shared information content of its gene pool.

Thus, the members of a species show the following properties:

(i) A reproductive community:

The member of an animal species recognises each other as potential mates and seek each other for the purpose of reproduction.

(ii) An ecological unit:

The species members form an ecological unit which, regardless of the individuals composing it, interacts as a unit with other species with which it shares the environment.

(iii) A genetic unit:

The species consists of a large, inter-communicating gene pool, whereas the individual is merely a temporary vessel holding a small portion of the contents of the gene pool for a short period of time.

These three properties raise the species above the typological interpretation of a “class of objects”. Thus, from this theoretical species concept, the species definition which results is — A species is a group of interbreeding natural population that is reproductively isolated from other such groups.

This species concept is called biological not because it deals with biological taxa, but because the definition itself is biological. It utilizes criteria that are meaningless as far as the inanimate world is concerned.

Biologically, a species is a potential gene pool. It is a Mendelian population which has its own devices, that is, isolating mechanisms which protect it against harmful gene flow from other gene pools. Gene of the same gene pool form harmonious combinations because they have become co-adopted by natural selection. Mixing the genes of two different species lead to high frequency of disharmonious gene combinations.

Mechanisms that prevent this are favoured by selection. Thus, the word species in biology is a relational term. ‘A’ is a species in relation to ‘B’ and ‘C’ because it is reproductively isolated from them. Since evolution is a regular process, the species is an arbitrary division of the continuous and ever-changing series of individuals in nature. Therefore, species is dynamic and multidimensional in nature.

The biological species concept has been able to solve the paradox caused by the conflict between the fixity of species of the naturalist and the fluidity of the species of the evolutionist. It was this conflict that made Linnaeus deny evolution and Darwin the reality of species.

The biological species concept combines the discreteness of the local species at a given time with an evolutionary potential for continuing change. The biological species concept has its importance in the fact that it is employed in the largest number of biological disciplines — ecology, physiology, behaviour biology etc.

Intraspecific categories designate groupings of population’s within species. Normally, the species is the lowest category used in routine taxonomy. The higher categories are groupings of species. In view of this key position of the species and the fact that in nature one encounters individuals and phena, the assigning of individuals and phena to species taxa is one of the key problems of taxonomy.

Q. Explain Cell Theory:

Microorganisms were unknown before the invention of improved microscopes, and human beings were believed to be the fundamental building blocks of life. Early scientists could observe cells and make assumptions about what they witnessed. It is rarely possible to know how a cell operates under a microscope.

A postulated and widely accepted hypothesis of how most life on Earth operates is known as the “cell theory.” The hypothesis states that cells comprise all organisms. Organs, tissues, and organisms are made up of groups of cells. These are the fundamental assumptions of cell theory.

The modern iteration of the cell theory consists of the following concepts:

- Energy moves through cells.
- DNA is transferred from cell to cell, carrying genetic information.
- All cells share the same primary chemical makeup.

In addition to cell theory, the fundamental precepts that serve as the basis for studying life include evolution, homeostasis, gene theory, and the laws of thermodynamics.

Q. Explain types of cell : Prokaryotic Cell and Eukaryotic Cell

Prokaryotic Cell :

“Prokaryotic cells are the cells that do not have a true nucleus and membrane-bound organelles.”

Prokaryotic cells are single-celled microorganisms known to be the earliest on earth. Prokaryotes include Bacteria and Archaea. The photosynthetic prokaryotes include cyanobacteria that perform photosynthesis.

A prokaryotic cell consists of a single membrane and therefore, all the reactions occur within the cytoplasm. They can be free-living or parasites.

Characteristics of Prokaryotic Cell

Prokaryotic cells have different characteristic features. The characteristics of the prokaryotic cells are mentioned below.

1. They lack a nuclear membrane.
2. Mitochondria, Golgi bodies, chloroplast, and lysosomes are absent.
3. The genetic material is present on a single chromosome.
4. The histone proteins, the important constituents of eukaryotic chromosomes, are lacking in them.
5. The cell wall is made up of carbohydrates and amino acids.
6. The plasma membrane acts as the mitochondrial membrane carrying respiratory enzymes.
7. They divide asexually by binary fission. The sexual mode of reproduction involves conjugation.

Prokaryotic Cell Structure

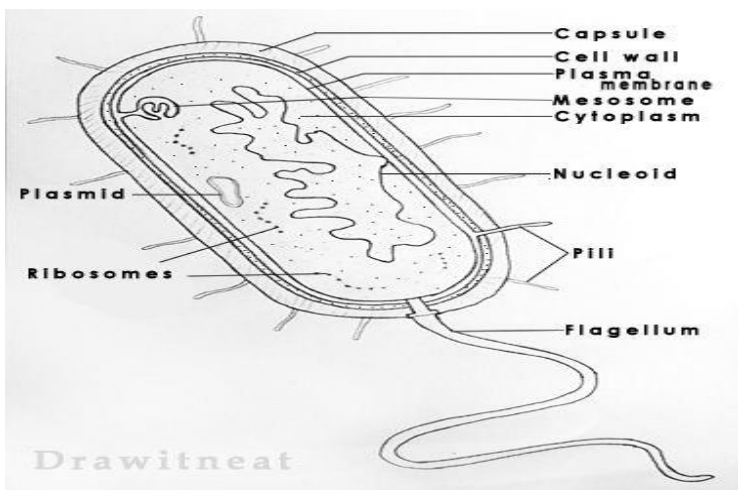
A prokaryotic cell does not have a nuclear membrane. However, the genetic material is present in a region in the cytoplasm known as the nucleoid. They may be spherical, rod-shaped, or spiral. A prokaryotic cell structure is as follows:

1. **Capsule**– It is an outer protective covering found in the bacterial cells, in addition to the cell wall. It helps in moisture retention, protects the cell when engulfed, and helps in the attachment of cells to nutrients and surfaces.
2. **Cell Wall**– It is the outermost layer of the cell which gives shape to the cell.
3. **Cytoplasm**– The cytoplasm is mainly composed of enzymes, salts, cell organelles and is a gel-like component.
4. **Cell Membrane**– This layer surrounds the cytoplasm and regulates the entry and exit of substances in the cells.
5. **Pili**– These are hair-like outgrowths that attach to the surface of other bacterial cells.
6. **Flagella**– These are long structures in the form of a whip, that help in the locomotion of a cell.
7. **Ribosomes**– These are involved in protein synthesis.
8. **Plasmids**– Plasmids are non-chromosomal DNA structures. These are not involved in reproduction.
9. **Nucleoid Region**– It is the region in the cytoplasm where the genetic material is present.

A prokaryotic cell lacks certain organelles like mitochondria, endoplasmic reticulum, and Golgi bodies.

Prokaryotic Cell Diagram

The prokaryotic cell diagram given below represents a bacterial cell. It depicts the absence of a true nucleus and the presence of a flagellum that differentiates it from a eukaryotic cell.



Eukaryotic Cell :

“Eukaryotic cells are the cells that contain a membrane bound nucleus and organelles.”

Eukaryotic cells have a nucleus enclosed within the nuclear membrane and form large and complex organisms. Protozoa, fungi, plants, and animals all have eukaryotic cells. They are classified under the kingdom Eukaryota.

They can maintain different environments in a single cell that allows them to carry out various metabolic reactions. This helps them grow many times larger than the prokaryotic cells.

The features of eukaryotic cells are as follows:

1. Eukaryotic cells have the nucleus enclosed within the nuclear membrane.
2. The cell has mitochondria.
3. Flagella and cilia are the locomotory organs in a eukaryotic cell.
4. A cell wall is the outermost layer of the eukaryotic cells.
5. The cells divide by a process called mitosis.
6. The eukaryotic cells contain a cytoskeletal structure.
7. The nucleus contains a single, linear DNA, which carries all the genetic information.

Structure Of Eukaryotic Cell: The eukaryotic cell structure comprises the following:

Plasma Membrane

- The plasma membrane separates the cell from the outside environment.
- It comprises specific embedded proteins, which help in the exchange of substances in and out of the cell.

Cell Wall

- A cell wall is a rigid structure present outside the plant cell. It is, however, absent in animal cells.
- It provides shape to the cell and helps in cell-to-cell interaction.
- It is a protective layer that protects the cell from any injury or pathogen attacks.
- It is composed of cellulose, hemicellulose, pectins, proteins, etc.

Cytoskeleton

The cytoskeleton is present inside the cytoplasm, which consists of microfilaments, microtubules, and fibres to provide perfect shape to the cell, anchor the organelles, and stimulate the cell movement.

Endoplasmic Reticulum

It is a network of small, tubular structures that divides the cell surface into two parts: luminal and extraluminal.

Endoplasmic Reticulum is of two types:

- Rough Endoplasmic Reticulum contains ribosomes.
- Smooth Endoplasmic Reticulum that lacks ribosomes and is therefore smooth.

Nucleus

- The nucleoplasm enclosed within the nucleus contains DNA and proteins.
- The nuclear envelop consists of two layers- the outer membrane and the inner membrane. Both the membranes are permeable to ions, molecules, and RNA material.
- Ribosome production also takes place inside the nucleus.

Golgi Apparatus

- It is made up of flat disc-shaped structures called cisternae.
- It is absent in red blood cells of humans and sieve cells of plants.
- They are arranged parallel and concentrically near the nucleus.
- It is an important site for the formation of glycoproteins and glycolipids.

Ribosomes

These are the main site for protein synthesis and are composed of proteins and ribonucleic acids.

Mitochondria

- These are also known as “powerhouse of cells” because they produce energy.
- It consists of an outer membrane and an inner membrane. The inner membrane is divided into folds called cristae.
- They help in the regulation of cell metabolism.

Lysosomes

They are known as “suicidal bags” because they possess hydrolytic enzymes to digest protein, lipids, carbohydrates, and nucleic acids.

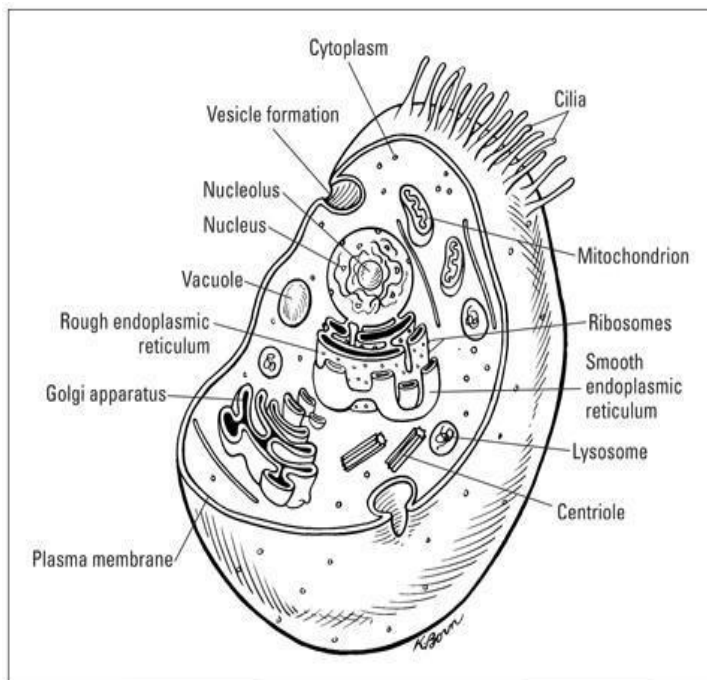
Plastids

These are double-membraned structures and are found only in plant cells. These are of three types:

- Chloroplast that contains chlorophyll and is involved in photosynthesis.
- Chromoplast that contains a pigment called carotene that provides the plants yellow, red, or orange colours.
- Leucoplasts that are colourless and store oil, fats, carbohydrates, or proteins.

Eukaryotic Cell Diagram

Eukaryotic cell diagram mentioned below depicts different cell organelles present in eukaryotic cells. The nucleus, endoplasmic reticulum, cytoplasm, mitochondria, ribosomes, lysosomes are clearly mentioned in the diagram.



What is the difference between Eukaryotic Cell and Prokaryotic Cell :

	Prokaryotes	Eukaryotes
Type of Cell	Always unicellular	Unicellular and multi-cellular
Cell size	Ranges in size from 0.2 μm – 2.0 μm in diameter	Size ranges from 10 μm – 100 μm in diameter
Cell wall	Usually present; chemically complex in nature	When present, chemically simple in nature
Nucleus	Absent. Instead, they have a nucleoid region in the cell	Present
Ribosomes	Present. Smaller in size and spherical in shape	Present. Comparatively larger in size and linear in shape
DNA arrangement	Circular	Linear
Mitochondria	Absent	Present

Cytoplasm	Present, but cell organelles absent	Present, cell organelles present
Endoplasmic reticulum	Absent	Present
Plasmids	Present	Very rarely found in eukaryotes
Ribosome	Small ribosomes	Large ribosomes
Lysosome	Lysosomes and centrosomes are absent	Lysosomes and centrosomes are present
Cell division	Through binary fission	Through mitosis
Flagella	The flagella are smaller in size	The flagella are larger in size
Reproduction	Asexual	Both asexual and sexual
Example	Bacteria and Archaea	Plant and Animal cell

Explain in detail Plant cell:

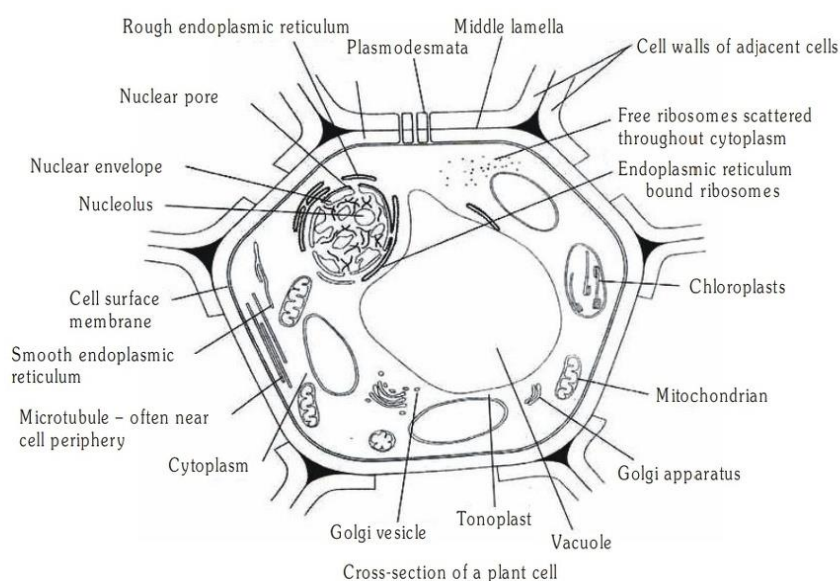
The cell is the basic unit of life in all organisms. Like humans and animals, plants are also composed of several cells. The plant cell is surrounded by a cell wall which is involved in providing shape to the plant cell. Apart from the cell wall, there are other organelles that are associated with different cellular activities.

Plant cells are eukaryotic cells that vary in several fundamental factors from other eukaryotic organisms. Both plant and animal cells contain a nucleus along with similar organelles. One of the distinctive aspects of a plant cell is the presence of a cell wall outside the cell membrane.

The plant cell is rectangular and comparatively larger than the animal cell. Even though plant and animal cells are eukaryotic and share a few cell organelles, plant cells are quite distinct when compared to animal cells as they perform different functions. Some of these differences can be clearly understood when the cells are examined under an electron microscope.

Plant Cell Structure

Just like different organs within the body, plant cell structure includes various components known as cell organelles that perform different functions to sustain itself. These organelles include:



Cell Wall

It is a rigid layer which is composed of polysaccharides cellulose, pectin and hemicellulose. It is located outside the cell membrane. It also comprises glycoproteins and polymers such as lignin, cutin, or suberin.

The primary function of the cell wall is to protect and provide structural support to the cell. The plant cell wall is also involved in protecting the cell against mechanical stress and providing form and structure to the cell. It also filters the molecules passing in and out of it.

The formation of the cell wall is guided by microtubules. It consists of three layers, namely, primary, secondary and the middle lamella. The primary cell wall is formed by cellulose laid down by enzymes.

Cell membrane

It is the semi-permeable membrane that is present within the cell wall. It is composed of a thin layer of protein and fat. The cell membrane plays an important role in regulating the entry and exit of specific substances within the cell. For instance, cell membrane keeps toxins from entering inside, while nutrients and essential minerals are transported across.

Nucleus

The nucleus is a membrane-bound structure that is present only in eukaryotic cells. The vital function of a nucleus is to store DNA or hereditary information required for cell division, metabolism and growth.

1. **Nucleolus:** It manufactures cells' protein-producing structures and ribosomes.
2. **Nucleopore:** Nuclear membrane is perforated with holes called nucleopore that allow proteins and nucleic acids to pass through.

Plastids

They are membrane-bound organelles that have their own DNA. They are necessary to store starch and to carry out the process of photosynthesis. It is also used in the synthesis of many molecules, which form the building blocks of the cell. Some of the vital types of **plastids** and their functions are stated below:

Leucoplasts

They are found in the non-photosynthetic tissue of plants. They are used for the storage of protein, lipid and starch.

Chloroplasts

It is an elongated organelle enclosed by phospholipid membrane. The chloroplast is shaped like a disc and the stroma is the fluid within the chloroplast that comprises a circular DNA. Each chloroplast contains a green coloured pigment called chlorophyll required for the process of photosynthesis. The chlorophyll absorbs light energy from the sun and uses it to transform carbon dioxide and water into glucose.

Chromoplasts

They are heterogeneous, coloured plastid which is responsible for pigment synthesis and for storage in photosynthetic eukaryotic organisms. Chromoplasts have red, orange and yellow coloured pigments which provide colour to all ripe fruits and flowers.

Central Vacuole

It occupies around 30% of the cell's volume in a mature plant cell. Tonoplast is a membrane that surrounds the central vacuole. The vital function of the central vacuole apart from storage is to sustain turgor pressure against the cell wall. The central vacuole consists of cell sap. It is a mixture of salts, enzymes and other substances.

Golgi Apparatus

They are found in all eukaryotic cells, which are involved in distributing synthesised macromolecules to various parts of the cell.

Ribosomes

They are the smallest membrane-bound organelles which comprise RNA and protein. They are the sites for protein synthesis, hence, also referred to as the protein factories of the cell.

Mitochondria

They are the double-membraned organelles found in the cytoplasm of all eukaryotic cells. They provide energy by breaking down carbohydrate and sugar molecules, hence they are also referred to as the "Powerhouse of the cell."

Lysosome

Lysosomes are called suicidal bags as they hold digestive enzymes in an enclosed membrane. They perform the function of cellular waste disposal by digesting worn-out organelles, food particles and foreign bodies in the cell. In plants, the role of lysosomes is undertaken by the vacuoles.

Animal Cell :

We are aware of the fact that the cell is the structural and fundamental unit of life. It is also the smallest and the most basic biological unit of living organisms. On the basis of the cellular organization, cells are further classified as eukaryotic and prokaryotic. Plant cells and animal cells fall under the eukaryotic category.

"An animal cell is a type of eukaryotic cell that lacks a cell wall and has a true, membrane-bound nucleus along with other cellular organelles."

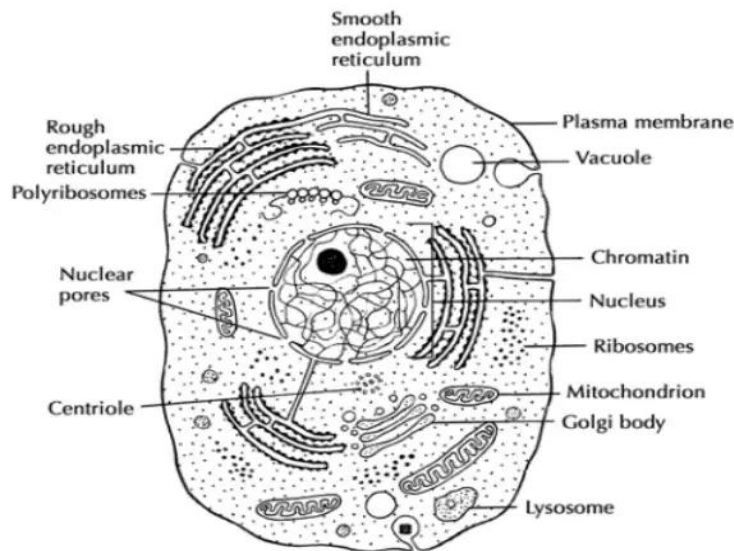
Animal cells range in size from a few microscopic microns to a few millimetres. The largest known animal cell is the ostrich egg, which can stretch over 5.1 inches across and weighs about 1.4 kilograms. This is in stark contrast to the neuron in the human body, which is just 100 microns across.

The shape of animal cells also varies, with some being flat, others oval or rod-shaped. There are also more intriguing shapes such as curved, spherical, concave and rectangular. Most of the cells are microscopic in size and can only be seen under the microscope.

As stated before, animal cells are eukaryotic cells with a membrane-bound nucleus. Furthermore, these cells exhibit the presence of DNA inside the nucleus. They also comprise other membrane-bound organelles and cellular structures which carry out specific functions necessary for a cell to function properly.

Animal Cell Diagram

The diagram given below depicts the structural organization of the animal cell. The various cell organelles present in an animal cell are clearly marked in the animal cell diagram provided below.



Though this animal cell diagram is not representative of any one particular type of cell, it provides insight into the primary organelles and the intricate internal structure of most animal cells. Furthermore, it is easy to distinguish between a plant and animal cell diagram just by inspecting the presence or absence of a cell wall.

Animal Cell Structure

Animal cells are generally smaller than plant cells. Another defining characteristic is its irregular shape. This is due to the absence of a cell wall. But animal cells share other cellular organelles with plant cells as both have evolved from eukaryotic cells.

A typical animal cell comprises the following cell organelles:

Cell Membrane

A thin semipermeable membrane layer of lipids and proteins surrounding the cell. Its primary role is to protect the cell from its surrounding. Also, it controls the entry and exit of nutrients and other microscopic entities into the cell. For this reason, cell membranes are known as semi-permeable or selectively permeable membranes.

Nucleus

It is an organelle that contains several other sub-organelles such as nucleolus, nucleosomes and chromatins. It also contains DNA and other genetic materials.

Nuclear Membrane

It is a double-membrane structure that surrounds the nucleus. It is also referred to as the nuclear envelope.

Centrosome

It is a small organelle found near the nucleus, which has a thick centre with radiating tubules. The centrosomes are where microtubules are produced.

Lysosome

They are round organelles surrounded by a membrane and comprising digestive enzymes which help in digestion, excretion and in the cell renewal process.

Cytoplasm

A jelly-like material which contains all the cell organelles, enclosed within the cell membrane. The substance found within the cell nucleus, contained by the nuclear membrane is called the nucleoplasm.

Golgi Apparatus

A flat, smooth layered, sac-like organelle which is located near the nucleus and involved in manufacturing, storing, packing and transporting the particles throughout the cell.

Mitochondrion

They are spherical or rod-shaped organelles with a double membrane. They are the powerhouse of a cell as they play an important role in releasing energy.

Ribosome

They are small organelles made up of RNA-rich cytoplasmic granules, and they are the sites of protein synthesis.

Endoplasmic Reticulum (ER)

This cellular organelle is composed of a thin, winding network of membranous sacs originating from the nucleus.

Vacuole

A membrane-bound organelle present inside a cell involved in maintaining shape and storing water, food, wastes, etc.

Nucleopore

They are tiny holes present in the nuclear membrane which are involved in the movement of nucleic acids and proteins across the nuclear membrane.

Write comparison between a plant cell and an animal cell:

S.No	Plant cell	Animal cell
1	10 to 100 µm in length.	10 to 30 µm in length.
2	Rectangular or cubic in shape.	Round or irregular shape.
3	Larger in size.	Smaller in size.
4	Cell wall is present.	Cell wall is absent.
5	Plastids are present.	Plastids are absent.
6	A large central vacuole is present.	Small vacuoles are present.
7	Golgi bodies found in the form of units viz., dictyosomes.	Golgi bodies are well developed.
8	Nucleus present at the peripheral side of the cytoplasm.	Nucleus lies down at the centre of the cell.
9	Chloroplast is present.	Chloroplast is absent.
10	Most of the plant cells don't have cilia.	Animal cells have cilia.
11	Mostly lysosomes do not present in the plant cell.	Lysosomes are present in the animal cell.
12	Plant cells store energy as starch.	Animal cells store energy as carbohydrate and glycogen.
13	Plasma membrane is hard and rigid.	Plasma membrane is thin and flexible.
14	Centrosomes and centrioles are absent.	Centrosomes and centrioles are present.
15	Plant cells can synthesize amino acids and vitamins.	Animal cells cannot synthesize amino acids and vitamins.
16	Plant cell contain fewer mitochondria.	Animal cell contain a huge number of mitochondria.

Q What is cell cycle?

A cell cycle is a series of events that takes place in a cell as it grows and divides. A cell spends most of its time in what is called interphase, and during this time it grows, replicates its chromosomes, and prepares for cell division. The cell then leaves interphase, undergoes mitosis, and completes its division. There are two types of cell division: mitosis and meiosis. Most of the time when people refer to "cell division," they mean mitosis, the process of making new body cells. Meiosis is the type of cell division that creates egg and sperm cells.

The cell cycle was discovered by Prevost and Dumas (1824) while studying the cleavage of zygote of Frog. It is a series of stages a cell passes through, to divide and produce new cells.

This entire process where with the help of one single parent cell a new cell population grows and develops is known as the cell cycle.

Phases of Cell Cycle

Cell cycle or cell division refers to the series of events that take place in a cell leading to its maturity and subsequent division. These events include duplication of its genome and synthesis of the cell organelles followed by division of the cytoplasm.

Human cells exhibit typical eukaryotic cell cycle and take around 24 hours to complete one cycle of growth and division. The duration of the cycle, however, varies from organism to organism and cell to cell.

A typical eukaryotic cell cycle is divided into two main phases:-

Interphase

Also known as the resting phase of the cell cycle; interphase is the time during which the cell prepares for division by undergoing both cell growth and DNA replication. It occupies around 95% time of the overall cycle. The interphase is divided into three phases:-

- **G1 phase (Gap 1)** – G1 phase is the phase of the cell between mitosis and initiation of replication of the genetic material of the cell. During this phase, the cell is metabolically active and continues to grow without replicating its DNA.
- **S phase (Synthesis)** – DNA replication takes place during this phase. If the initial quantity of DNA in the cell is denoted as $2N$, then after replication it becomes $4N$. However the number of chromosomes does not vary, viz., if the number of chromosomes during G1 phase was $2n$, it will remain $2n$ at the end of S phase. The centriole also divides into two centriole pairs in the cells which contain centriole.
- **G2 phase (Gap 2)** – During this phase, the RNA, proteins, other macromolecules required for multiplication of cell organelles, spindle formation, and cell growth are produced as the cell prepares to go into the mitotic phase.

Some cells like cardiac cells in the adult animals do not exhibit division and some others only divide to replace those cells which have been either damaged or lost due to cell death. Such cells which do not divide further attain an inactive G0 phase also known as quiescent phase after they exit the G1 phase. These cells remain metabolically active but do not divide unless called upon to do so.

M phase

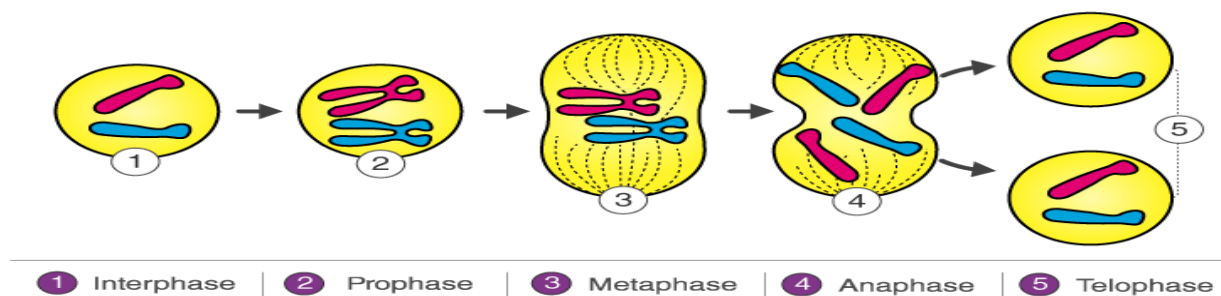
This is the mitotic phase or the phase of the equational division as the cell undergoes a complete reorganization to give birth to a progeny that has the same number of chromosomes as the parent cell. The other organelles are also divided equally by the process of cytokinesis which is preceded by mitotic nuclear division. The mitotic phase is divided into four overlapping stages:-

1. Prophase,
2. Metaphase,
3. Anaphase, and
4. Telophase

The stages of Mitosis are:

- Prophase – The chromosomes shorten and thicken.
- Metaphase – Chromosomes line up in the middle of the cell.
- Anaphase – Chromatids break apart at the centromere and move to opposite poles.
- Telophase – Two nuclei formed after nuclear envelopes reform around each group of chromosomes.

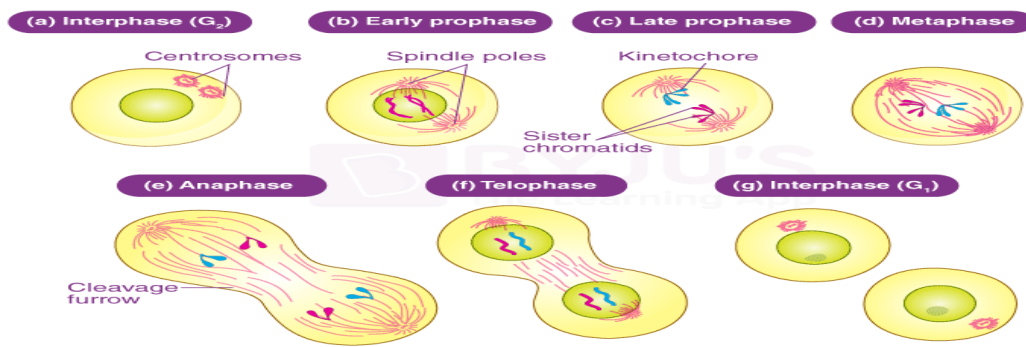
Mitosis



The process by which a eukaryotic cell separates the nuclear DNA and chromosomes and divides into two different but similar sets of nuclei is known as mitosis. The chromosomes are pulled apart by a mitotic spindle, which is a specialized structure consisting of microtubules.

Cytokinesis

In this phase, the cytoplasm of the cell divides. It begins as soon as the mitosis ends. Plant cells are much tougher than animal cells, as they have a rigid cell wall and high internal pressure. Thus, cytokinesis occurs in plant and animal cells differently. Cell division happens when a parent cell divides into two or more cells called daughter cells. Cell division usually occurs as part of a larger cell cycle. All cells reproduce by splitting into two, where each parental cell gives rise to two daughter cells.



These newly formed daughter cells could themselves divide and grow, giving rise to a new cell population that is formed by the division and growth of a single parental cell and its descendant.

In other words, such cycles of growth and division allow a single cell to form a structure consisting of millions of cells.

Explain Mitosis and Meiosis :

- Mitosis is a continuous process of cell division which occurs in all types of living cells.
- Mitosis involves four basic phases – prophase, metaphase, anaphase and telophase.
- Mitosis is the process where the division of cell occurs by asexual reproduction.
- In mitosis, the nuclear membrane is broken down, spindle fibres (microtubules) attach to the chromatids at the centromere and pull apart the chromatids.
- When the chromatids reach separate ends of the cells, the spindle fibres disintegrate and a nuclear membrane rebuilds around the chromosomes making two nuclei.
- Each nucleus is identical to the original nucleus as it was in G₁ phase.

Meiosis

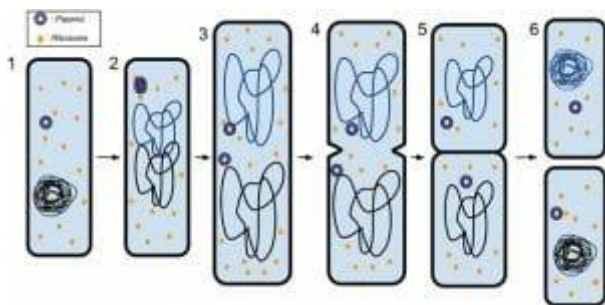
- Meiosis is the form of nuclear cell division that results in daughter cells that have one-half the chromosome numbers as the original cell.
- In organisms that are diploid, the end result is cells that are haploid. Each daughter cell gets one complete set of chromosomes, i.e., one of each homologous pair of chromosomes.
- In humans, this means the chromosome number is reduced from 46 to 23.
- The germ cells undergo meiosis to give rise to sperm and eggs.
- The joining together of a sperm and egg during fertilisation returns the number of the chromosomes to 46.
- Cells that undergo meiosis go through the cell cycle, including the S phase, so the process begins with chromosomes that consist of two chromatids just as in mitosis.
- Meiosis consists of meiosis I and meiosis II. In meiosis I, homologous chromosomes are separated into different nuclei.
- This is the reduction division; chromosome number is divided in half. Meiosis II is very similar to mitosis; chromatids are separated into separate nuclei.
- As in mitosis, it is spindle fibres that “pull” the chromosomes and chromatids apart in meiosis.
- The end result of meiosis is four cells, each with one complete set of chromosomes instead of two sets of chromosomes.

Q. Explain differences between Mitosis and Meiosis are as follows:

Types of Cell Division:

Prokaryotic Cell Division

Prokaryotes replicate through a type of cell division known as *binary fission*. Prokaryotes are simple organism, with only one membrane and no division internally. Thus, when a prokaryote divides, it simply replicates the DNA and splits in half. The process is a little more complicated than this, as DNA must first be unwound by special proteins. Although the DNA in prokaryotes usually exists in a ring, it can get quite tangled when it is being used by the cell. To copy the DNA efficiently, it must be stretched out. This also allows the two new rings of DNA created to be separated after they are produced. The two strands of DNA separate into two different sides of the prokaryote cell. The cell then gets longer, and divides in the middle. The process can be seen in the image below.

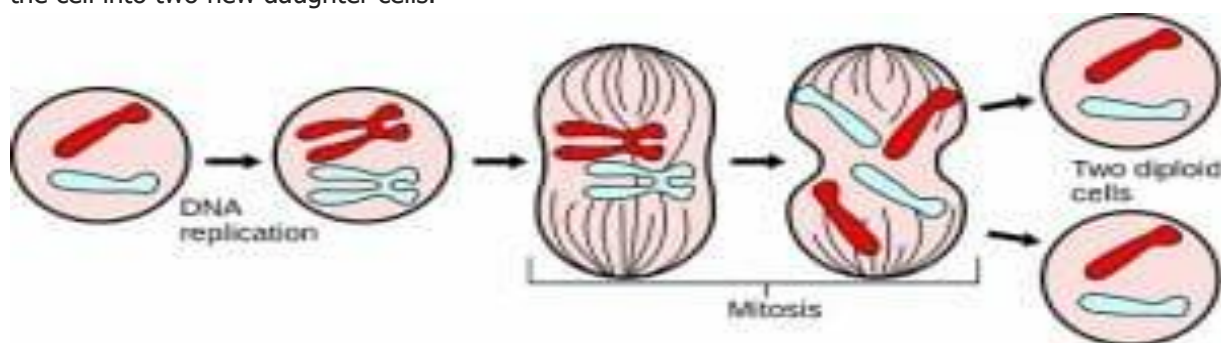


The DNA is the tangled line. The other components are labeled. Plasmids are small rings of DNA that also get copied during *binary fission* and can be picked up in the environment, from dead cells that break apart. These plasmids can then be further replicated. If a plasmid is beneficial, it will increase in a population. This is in part how antibiotic resistance in bacteria happens. The ribosomes are small protein structures that help produce proteins. They are also replicated so each cell can have enough to function.

Eukaryotic Cell Division: Mitosis

Eukaryotic organisms have membrane bound organelles and DNA that exists on chromosomes, which makes cell division harder. Eukaryotes must replicate their DNA, organelles, and cell mechanisms before dividing. Many of the organelles divide using a process that is essentially *binary fission*, leading scientist to believe that eukaryotes were formed by prokaryotes living inside of other prokaryotes.

After the DNA and organelles are replicated during *interphase* of the cell cycle, the eukaryote can begin the process of mitosis. The process begins during *prophase*, when the chromosomes condense. If mitosis proceeded without the chromosomes condensing, the DNA would become tangled and break. Eukaryotic DNA is associated with many proteins which can fold it into complex structures. As mitosis proceeds to *metaphase* the chromosomes are lined up in the middle of the cell. Each half of a chromosome, known as *sister chromatids* because they are replicated copies of each other, gets separated into each half of the cell as mitosis proceeds. At the end of mitosis, another process called *cytokinesis* divides the cell into two new daughter cells.



All eukaryotic organisms use mitosis to divide their cells. However, only single-celled organisms use mitosis as a form of reproduction. Most multicellular organisms are sexually reproducing and combine their DNA with that of another organism to reproduce. In these cases, organisms need a different method of cell division. Mitosis yields identical cells, but meiosis produces cells with half the genetic information of a regular cell, allowing two cells from different organisms of the same species to combine.

Eukaryotic Cell Division: Meiosis

In sexually reproducing animals, it is usually necessary to reduce the genetic information before fertilization. Some plants can exist with too many copies of the genetic code, but in most organisms it is highly detrimental to have too many copies. Humans with even one extra copy of one chromosome can experience detrimental changes to their body. To counteract this, sexually reproducing organisms undergo a type of cell division known as *meiosis*. As before mitosis, the DNA and organelles are replicated. The process of meiosis contains two different cell divisions, which happen back-to-back. The first meiosis, *meiosis I*, separates homologous chromosomes. The homologous chromosomes present in a cell represent the two alleles of each gene an organism has. These alleles are recombined and separated, so the resulting daughter cells have only one allele for each gene, and no homologous pairs of chromosomes. The second division, *meiosis II*, separated the two copies of DNA, much like in mitosis. The end result of meiosis in one cell is 4 cells, each with only one copy of the genome, which is half the normal number.

Organisms typically package these cells into *gametes*, which can travel into the environment to find other gametes. When two gametes of the right type meet, one will fertilize the other and produce a *zygote*. The zygote is a single cell that will undergo mitosis to produce the millions of cells necessary for a large organism. Thus, most eukaryotes use both mitosis and meiosis, but at different stages of their lifecycle.

Cell Division Stages

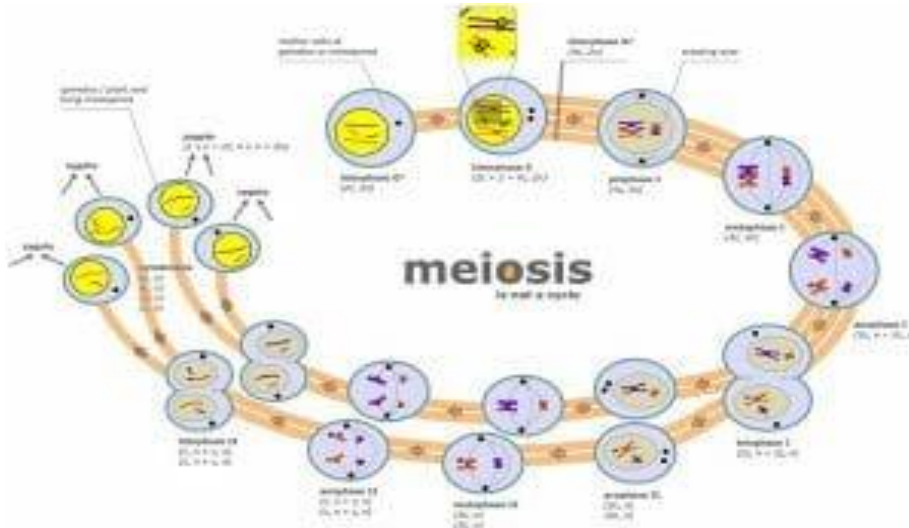
Depending upon which type of cell division an organism uses, the stages can be slightly different.

Mitosis Stages

Mitosis starts with *prophase* in which the chromosome is condensed. The cell proceeds to *metaphase* where the chromosomes are aligned on the metaphase plate. Then the chromosomes are separated in *anaphase* and the cell's cytoplasm is pinched apart during *telophase*. *Cytokinesis* is the final process that breaks the cell membrane and divides the cell into two.

Meiosis Stages

The stages of meiosis are similar to mitosis, but the chromosomes act differently. Meiosis has two phases, which include two separate cell divisions without the DNA replicating between them. *Meiosis I* and *meiosis II* have the same 4 stages as mitosis: prophase, metaphase, anaphase, and telophase. Cytokinesis concludes both rounds of meiosis.



In prophase I, the chromosomes are condensed. In metaphase I, the chromosomes line up across from their homologous pairs. When they are separated in anaphase I and telophase I, there is only one form of each gene in each cell, known as a reduction division. Meiosis II proceeds in the same manner as mitosis, which sister chromatids dividing on the metaphase plate. By telophase II, there are 4 cells, each with half of the alleles as the parent cell and only a single copy of the genome. The cells can now become gametes and fuse together to create new organisms.

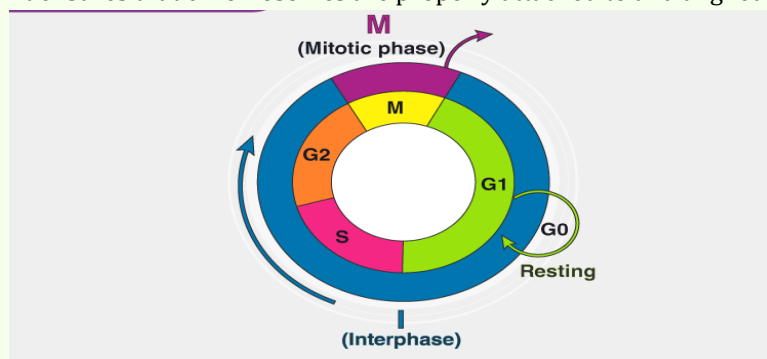
Q. What are checkpoints in the cell cycle?

Cell cycle:

1. It is a set of events that occur in a cell when it divides and grows.
2. It is divided into two phases - the Interphase and the M phase.

Cell cycle checkpoints:

1. Cell cycle checkpoints are control mechanisms that ensure the normal course of the eukaryotic cell cycle.
2. In the cell cycle, there are three major checkpoints - the G_1 checkpoint, the G_2 checkpoint, and the spindle checkpoint.
3. At the G_1/S transition, the G_1 checkpoint occurs.
4. At the G_1 checkpoint, it's determined whether the cell is large enough to divide, has enough nourishment for cell division, and is free of DNA damage.
5. At the G_2/M transition, the G_2 checkpoint occurs.
6. It is assessed whether the DNA has been completely duplicated at the G_2 checkpoint.
7. At the transition from metaphase to anaphase, the spindle checkpoint occurs.
8. The spindle checkpoint ensures that chromosomes are properly attached to and aligned on the spindle.



Q. Explain in detail Cell Chemistry :

Biomolecules

Biomolecules are biological molecules produced by the cells of the living organism. They are critical for life as it helps organisms to carry out basic biological processes such as reproduction, growth and sustenance. Biomolecules include both micromolecules, e.g. amino acids, nitrogenous bases, fatty acids, sugar, etc. and macromolecules, such as carbohydrates, proteins, lipids and nucleic acids. Living organisms are made up of organic as well as inorganic substances.

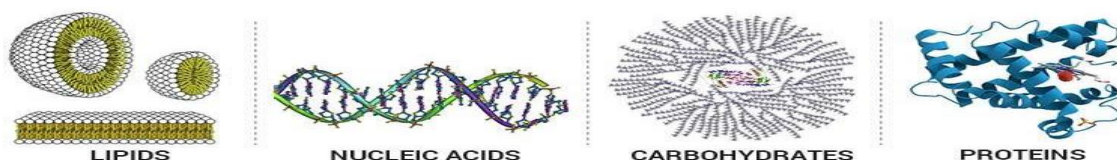
All the macromolecules are homo or hetero polymers of various simple compounds. Protein- Polymer of amino acids

Carbohydrates (Polysaccharides)- Polymer of simple sugars, e.g. glucose, fructose

Fats- Fatty acids and glycerol

Nucleic acids- Nitrogenous bases, sugar and phosphate

Water constitutes 70-90% of cells. Proteins are 10-15%, nucleic acids are 5-7%, carbohydrates are 3%, lipids are 2% and the rest 1% are ions.



There are four major classes of Biomolecules – Carbohydrates, Proteins, Nucleic acids and Lipids. Each of them is discussed below.

Carbohydrates

Carbohydrates are macronutrients and are one of the three main ways by which our body obtains its energy. They are called carbohydrates as they comprise carbon, hydrogen and oxygen at their chemical level. Carbohydrates are essential nutrients which include sugars, fibers and starches. They are found in grains, vegetables, fruits and in milk and other dairy products. They are the basic food groups which play an important role in a healthy life.

The food containing carbohydrates are converted into glucose or blood sugar during the process of digestion by the digestive system.

Our body utilizes this sugar as a source of energy for the cells, organs and tissues. The extra amount of energy or sugar is stored in our muscles and liver for further requirement. The term 'carbohydrate' is derived from a French term 'hydrate de carbone' meaning 'hydrate of carbon'. The general formula of this class of organic compounds is $C_n(H_2O)_n$.

Classification of Carbohydrates

The carbohydrates are further classified into simple and complex which is mainly based on their chemical structure and degree of polymerization.

Simple Carbohydrates (Monosaccharides, Disaccharides and Oligosaccharides)

Simple carbohydrates have one or two sugar molecules. In simple carbohydrates, molecules are digested and converted quickly resulting in a rise in the blood sugar levels. They are abundantly found in milk products, beer, fruits, refined sugars, candies, etc.

Proteins

Proteins are another class of indispensable biomolecules, which make up around 50 per cent of the cellular dry weight. Proteins are polymers of **amino acids** arranged in the form of polypeptide chains. The structure of proteins is classified as primary, secondary, tertiary and quaternary in some cases. These structures are based on the level of complexity of the folding of a polypeptide chain. Proteins play both structural and dynamic roles. Myosin is the protein that allows movement by contraction of muscles. Most enzymes are proteinaceous in nature.

Nucleic Acids

Nucleic acids refer to the genetic material found in the cell that carries all the hereditary information from parents to progeny. There are two types of nucleic acids namely, deoxyribonucleic acid (DNA) and ribonucleic acid (RNA). The main function of nucleic acid is the transfer of genetic information and synthesis of proteins by processes known as translation and transcription. The monomeric unit of nucleic acids is known as nucleotide and is composed of a nitrogenous base, pentose sugar, and phosphate. The nucleotides are linked by a 3' and 5' phosphodiester bond. The nitrogen base attached to the pentose sugar makes the nucleotide distinct. There are 4 major nitrogenous bases found in DNA: adenine, guanine, cytosine, and thymine. In RNA, thymine is replaced by uracil. The DNA structure is described as a double-helix or double-helical structure which is formed by hydrogen bonding between the bases of two antiparallel polynucleotide chains. Overall, the **DNA structure** looks similar to a twisted ladder.

Lipids

Lipids are organic substances that are insoluble in water, soluble in organic solvents, are related to fatty acids and are utilized by the living cell. They include fats, waxes, sterols, fat-soluble vitamins, mono-, di- or triglycerides, phospholipids, etc. Unlike carbohydrates, proteins, and nucleic acids, lipids are not polymeric molecules. Lipids play a great role in the cellular structure and are the chief source of energy.

Explain Cell death:

Cell death refers to the event that leads to the death of a cell. The process entails the breaking down of cell structures. The death of a cell may be determined when it has completed a fixed number of division cycles or at some earlier stage when programmed to do so, as in digit separation in vertebrate limb morphogenesis.

Cell death has two major kinds: (1) programmed cell death and (2) necrosis. Programmed cell death is a kind of cell death that is regulated by an innate physiological system. The cell undergoes a natural process of death after completing a number of cell divisions. The cell ceases to carry out its functions and dies to be replaced by new ones. The programmed type of cell death is further categorized into type I cell death or apoptosis and type II cell death or autophagy (self-digestion). In contrast, necrosis is the death of the cell due to factors such as disease, injury, or the death of the organism.

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